Formal Proof: Predictive Risk in High-Dimensional Transfer Learning

1. Notation and Regularity Conditions

In addition to the High-Dimensional Assumption and Random Regression Coefficients Assumption, we formally establish the following regularity condition to ensure the existence of limiting spectral distributions.

Assumption 1 (Covariance Regularity): The population covariance matrix $\Sigma \in \mathbb{R}^{p\times p}$ is strictly positive definite. As $p \to \infty$, the spectral norm is uniformly bounded, $c \leq \lambda_{\min}(\Sigma) \leq \lambda_{\max}(\Sigma) \leq C$ for some absolute constants $0 < c < C < \infty$.

Notation: Let $X^{(m)}_{\mathcal{R}(m)} \in \mathbb{R}^{n_m \times p_m}$ be the observed design matrix at site m. The true conditional coefficient on the observed subspace is $\omega^{(m)} = \Phi^{(m)}\beta^{(m)}$. The effective noise at site m is $\tilde{\varepsilon}^{(m)} = Y^{(m)} - X^{(m)}_{\mathcal{R}(m)}\omega^{(m)}$, which incorporates both the intrinsic noise ε and the variation from the unobserved covariates. The variance of this effective noise is $\tilde{\sigma}^2_m = \sigma^2 + \frac{\alpha^2_m}{p}\mathrm{Tr}(\Sigma_{[\mathcal{R}(m)]^c|\mathcal{R}(m)})$.

2. Step 1: Local Ridge Estimation and Deterministic Equivalents

At each source site $m \in \{1, \ldots, M\}$, the local ridge estimator is:

$$\widehat{\omega}^{(m)}_{\mathrm{Loc}} = \hat{\Sigma}^{-1}_{m,\lambda}\frac{1}{n_m}(X^{(m)}_{\mathcal{R}(m)})^\top Y^{(m)}$$

where $\hat{\Sigma}_{m,\lambda} = \frac{1}{n_m}(X^{(m)}_{\mathcal{R}(m)})^\top X^{(m)}_{\mathcal{R}(m)} + \lambda^{(m)}_\omega I_{p_m}$.

Using the techniques from **Dobriban & Wager (2018)**, as $n_m, p_m \to \infty$ with $p_m/n_m \to \eta_m$, the resolvent matrix $\hat{\Sigma}^{-1}_{m,\lambda}$ admits a deterministic equivalent.

The estimation error at site m can be decomposed as:

$$\Delta^{(m)}_{\mathrm{Loc}} := \widehat{\omega}^{(m)}_{\mathrm{Loc}} - \omega^{(m)} = -\lambda^{(m)}_\omega \hat{\Sigma}^{-1}_{m,\lambda}\omega^{(m)} + \hat{\Sigma}^{-1}_{m,\lambda}\frac{1}{n_m}(X^{(m)}_{\mathcal{R}(m)})^\top \tilde{\varepsilon}^{(m)}$$

Under the random coefficients assumption, $\omega^{(m)} \sim N\left(0, \frac{\alpha^2_m}{p}\Phi^{(m)}(\Phi^{(m)})^\top\right)$. The asymptotic covariance of the local estimation error, conditional on the true parameter but marginalized over the data generation process, converges to a deterministic matrix Ω_m:

$$\Omega_m := \lim_{p\to\infty} \mathbb{E}\left[\Delta^{(m)}_{\mathrm{Loc}}(\Delta^{(m)}_{\mathrm{Loc}})^\top\right] = B_m + V_m$$

where B_m is the bias covariance and V_m is the variance covariance. By random matrix theory, B_m and V_m are bounded and can be characterized by the limiting Stieltjes transform of the empirical spectral distribution of $\hat{\Sigma}_{m,0}$.

3. Step 2: Aggregation and Optimal Weighting Metrics

Following **Cheng, Cheng, & Duchi (2023)**, the aggregator targets the full-feature vector $\beta^{(t)}$ by solving:

$$\hat{\beta}^{(s)} = \arg\min_\beta \sum_{m=1}^M \left\|\Phi^{(m)}\beta - \widehat{\omega}^{(m)}_{\mathrm{Loc}}\right\|^2_{W_m}$$

This generalized least squares problem has the closed-form solution:

$$\hat{\beta}^{(s)} = \left(\sum_{m=1}^M (\Phi^{(m)})^\top W_m \Phi^{(m)}\right)^{-1} \sum_{m=1}^M (\Phi^{(m)})^\top W_m \widehat{\omega}^{(m)}_{\mathrm{Loc}}$$

To determine the optimal W_m, we analyze the deviation from the target site's true coefficient $\beta^{(t)}$:

$$\widehat{\omega}^{(m)}_{\mathrm{Loc}} - \Phi^{(m)}\beta^{(t)} = \Delta^{(m)}_{\mathrm{Loc}} + \Phi^{(m)}(\beta^{(m)} - \beta^{(t)})$$

By the Random Regression Coefficients Assumption, the covariance of the distributional shift $\beta^{(m)} - \beta^{(t)}$ is:

$$\mathrm{Cov}(\beta^{(m)} - \beta^{(t)}) = \frac{\alpha_m^2 + \alpha_t^2 - 2\rho_{m,t}\alpha_m\alpha_t}{p} I_p := \frac{\tau_m^2}{p} I_p$$

Assuming independence of source data, the total effective variance of the local estimator relative to the target parameter $\beta^{(t)}$ is:

$$\Gamma_m := \Omega_m + \frac{\tau_m^2}{p} \Phi^{(m)} (\Phi^{(m)})^\top$$

By the Gauss-Markov theorem in the Loewner ordering, the aggregation error $\Sigma_{\text{agg}} = \mathbb{E}[(\hat{\beta}^{(s)} - \beta^{(t)})(\hat{\beta}^{(s)} - \beta^{(t)})^\top]$ is minimized when the weights are chosen as the inverse of the effective variance:

$$W_m^* = \Gamma_m^{-1} = \left(\Omega_m + \frac{\tau_m^2}{p} \Phi^{(m)} (\Phi^{(m)})^\top \right)^{-1}$$

With this optimal weighting, the asymptotic error covariance of the aggregated vector is:

$$\Sigma_{\text{agg}} = \left(\sum_{m=1}^{M} (\Phi^{(m)})^\top (W_m^*) \Phi^{(m)} \right)^{-1}$$

4. Step 3: L_2 Transfer Learning and Predictive Risk Decomposition

Let the offset be $\hat{\omega}_s^{(t)} = \Phi^{(t)} \hat{\beta}^{(s)}$. The target L_2 transfer learning estimator estimates the shift $\delta = \omega^{(t)} - \hat{\omega}_s^{(t)}$:

$$\hat{\delta} = \arg\min_{\delta} \left[\frac{1}{n_t} \| Y^{(t)} - X_{\mathcal{R}(t)}^{(t)} (\hat{\omega}_s^{(t)} + \delta) \|_2^2 + \lambda_\delta \|\delta\|_2^2 \right]$$

$$\hat{\omega}_{\text{TL}}^{(t)} = \hat{\omega}_s^{(t)} + \hat{\delta}$$

Following the decomposition strategy from **Gu, Han, and Duan (2025)**, we express the prediction error on a new sample $\tilde{x}_{\mathcal{R}(t)}$. Let $H_\delta = \frac{1}{n_t} (X_{\mathcal{R}(t)}^{(t)})^\top X_{\mathcal{R}(t)}^{(t)} + \lambda_\delta I_{p_t}$. The shift estimator is:

$$\hat{\delta} = H_\delta^{-1} \frac{1}{n_t} (X_{\mathcal{R}(t)}^{(t)})^\top (Y^{(t)} - X_{\mathcal{R}(t)}^{(t)} \hat{\omega}_s^{(t)})$$

Substituting $Y^{(t)} = X_{\mathcal{R}(t)}^{(t)} \omega^{(t)} + \tilde{\varepsilon}^{(t)}$, the gap between the TL estimator and the true conditional coefficient is:

$$\hat{\omega}_{\text{TL}}^{(t)} - \omega^{(t)} = \lambda_\delta H_\delta^{-1} (\hat{\omega}_s^{(t)} - \omega^{(t)}) + H_\delta^{-1} \frac{1}{n_t} (X_{\mathcal{R}(t)}^{(t)})^\top \tilde{\varepsilon}^{(t)}$$

The predictive risk r^{pred} evaluates the out-of-sample expected squared error. Taking the expectation over the target site data and the prior aggregation, the risk decomposes into:

$$\mathbb{E}[r^{\text{pred}}(\hat{\omega}_{\text{TL}}^{(t)})] \simeq \tilde{\sigma}_t^2 + \mathcal{B}_{\text{TL}} + \mathcal{V}_{\text{TL}}$$

Where:

1. **Irreducible Error:** $\tilde{\sigma}_t^2 = \sigma^2 + \frac{\alpha_t^2}{p} \mathrm{Tr}(\Sigma_{[\mathcal{R}(t)]^c | \mathcal{R}(t)})$.
2. **Bias (Transferred Aggregation Error):**

$$\mathcal{B}_{\text{TL}} = \lambda_\delta^2 \, \mathrm{Tr} \left(\Sigma_{\mathcal{R}(t),\mathcal{R}(t)} H_\delta^{-1} \Phi^{(t)} \Sigma_{\text{agg}} (\Phi^{(t)})^\top H_\delta^{-1} \right)$$

3. **Variance (Target Domain Fine-Tuning):**

$$\mathcal{V}_{\text{TL}} = \frac{\tilde{\sigma}_t^2}{n_t} \mathrm{Tr} \left(\Sigma_{\mathcal{R}(t),\mathcal{R}(t)} H_\delta^{-1} \hat{\Sigma}_{t,0} H_\delta^{-1} \right)$$

5. Final Asymptotic Upper Bound

As $n_t, p \to \infty$ with $p_t/n_t \to \eta_t$, we define the limiting constants via deterministic equivalents:

$$v_t = \lim_{p\to\infty} \frac{1}{p_t} \mathrm{Tr}\left(\Sigma_{\mathcal{R}(t),\mathcal{R}(t)} H_\delta^{-1} \hat{\Sigma}_{t,0} H_\delta^{-1}\right)$$

$$b_t(\Sigma_{\mathrm{agg}}) = \lim_{p\to\infty} \frac{1}{p_t} \mathrm{Tr}\left(\Sigma_{\mathcal{R}(t),\mathcal{R}(t)} H_\delta^{-1} \Phi^{(t)} \Sigma_{\mathrm{agg}} (\Phi^{(t)})^\top H_\delta^{-1}\right)$$

Main Bound: Under the stated assumptions, as $n_t, p \to \infty$, the predictive risk of the transfer learning estimator converges in probability to the asymptotic upper bound:

$$\limsup_{n_t,p\to\infty} r^{\mathrm{pred}}\left(\hat{\omega}_{\mathrm{TL}}^{(t)}\right) = \tilde{\sigma}_t^2 + \lambda_\delta^2 \cdot \eta_t \cdot b_t(\Sigma_{\mathrm{agg}}) + \frac{\tilde{\sigma}_t^2 \cdot \eta_t \cdot v_t}{\lambda_\delta}$$